Perceiver & Perceiver IO

Implementation and Experimental Analysis

Francesco Gigli e Corso Vignoli

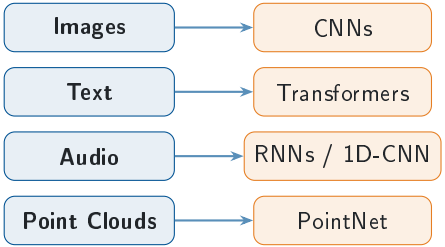
Università degli Studi di Firenze
Course B031278 — Deep Learning

Agenda

1. Introduction & Motivation
2. Perceiver: Architecture, FFN, Weight Sharing
3. Perceiver: Experiments & Results
4. Perceiver IO: Decoder, Queries, Results

The Problem: Modality-Specific Architectures

One modality, one architecture



Each data type is usually paired with a dedicated model and domain-specific assumptions about structure.

The scaling bottleneck

Self-attention cost

$$\mathcal{O}(M^2 \cdot d)$$

grows quadratically with input length M

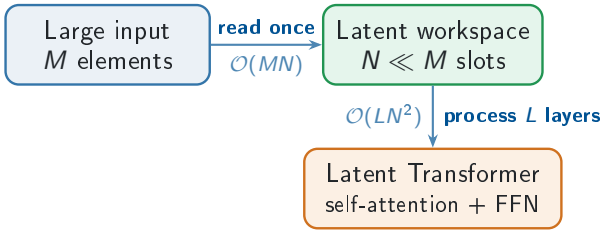
Input representation	M
16×16 image patches	256
raw 224×224 pixels	50,176

Raw pixels make standard attention impractical.

Core question: can one architecture handle arbitrary inputs without quadratic cost?

The Perceiver Idea: Read Big, Think Small

Compress first: map M raw input elements into $N \ll M$ learned latents, then run the deep Transformer only on that compact workspace.

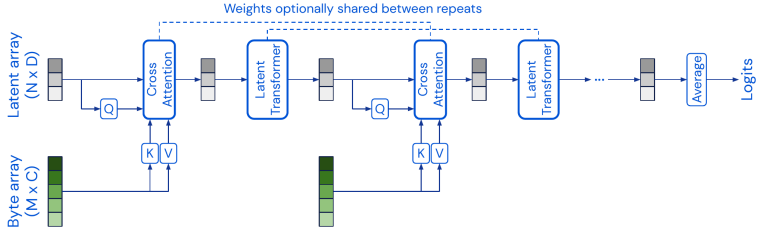


Perceiver cost $= \mathcal{O}(MN + LN^2)$

What changes?

- No self-attention over the full raw sequence.
- Repeated computation depends on N, not on M.
- The input can be pixels, tokens, points, or bytes.

Perceiver Architecture



Architecture flow

- Input array X is read by N learned latents L.
- Cross-attention is the only direct input read: $\mathcal{O}(MN)$.
- Latent Transformer repeats self-attention + FFN; weights can be shared.

Original Perceiver output

- Mean-pool the final latents.
- Linear classifier produces logits.
- No output queries; Perceiver IO adds them later.

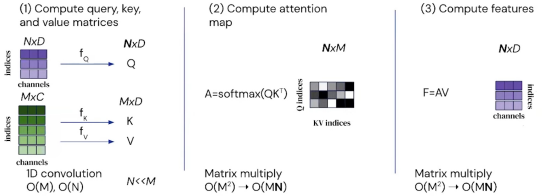
Cross-Attention: The Key Mechanism

Formally:

$$Q = X_{\text{latent}}\, W_Q \in \mathbb{R}^{N \times d_k}$$
$$K = X_{\text{input}}\, W_K \in \mathbb{R}^{M \times d_k}$$
$$V = X_{\text{input}}\, W_V \in \mathbb{R}^{M \times d_v}$$

$$\text{CrossAttn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

The latents are the queries, the input provides keys and values $\Rightarrow$ informational *bottleneck*.



Key insight:

- Attention matrix: $N \times M$ (not $M \times M$)
- Complexity: $\mathcal{O}(N \cdot M \cdot d_k)$
- Decouples depth from input size

Latent Transformer & Weight Sharing

Self-Attention among latents:

$$Q = K = V = X_{\text{latent}} \quad \Rightarrow \quad \mathcal{O}(N^2)$$

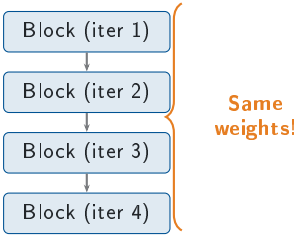
- Pre-norm LayerNorm
- Residual connections
- Feed-forward with $4\times$ expansion
- GEGLU activation (gated linear units)

GEGLU:

$$\text{GEGLU}(x) = (xW_1) \odot \text{GELU}(xW_2)$$

Higher non-linear capacity, stable gradient flow.

Weight sharing mechanism:



Configuration	Parameters
Unshared ($T = 4$ iters)	$\sim$8.67M
Shared (T iterations)	**$\sim$3.35M**
Reduction	$\sim 2.6\times$

Fourier Positional Encoding

Why is it needed?

- Attention is *permutation-equivariant*
- Without PE the model has no notion of "where" data is
- Experimentally confirmed: 61.34% without PE vs 72.02% with PE

Fourier PE (per axis x_d, raw coord. concatenated):

$$\gamma(x_d) = \left[x_d,\ \sin(f_k \pi x_d),\ \cos(f_k \pi x_d) \right]_{k=1}^{K}$$

$K = 64$ frequencies **linearly spaced** in $[1,\ \mu/2]$ ($\mu/2$ = Nyquist).

Resulting dimensions:

Modality	Pos. features	C_{tot}
Image (2D)	$3 + 2(2K{+}1)$	261
Points (3D)	$3 + 3(2K{+}1)$	390
Text (byte)	$128 + 64$	192

Critical Finding

Positional encoding is the **single most important** component: removing it costs -10.68**pp** (72.02% $\rightarrow$ 61.34%).

Perceiver Forward Pass: Shape Checkpoints

Data flow in the original Perceiver:

1. Input is flattened and enriched:

$$X = [\text{content} \,\|\, \gamma(\text{position})] \in \mathbb{R}^{M \times C_{\text{tot}}}$$

2. Learned latent array:

$$L_0 \in \mathbb{R}^{N \times D}, \qquad N \ll M$$

3. Cross-attention reads input:

$$Q = LW_Q, \quad K = XW_K, \quad V = XW_V$$

4. Latent Transformer applies self-attention + FFN only on N latents.

ImageNet reference shapes:

Object	Shape	Meaning
X	$50{,}176 \times 261$	pixels + Fourier PE
L_0	512×1024	learned latents
Q	512×261	latent queries
K, V	$50{,}176 \times 261$	input keys/values
A	$512 \times 50{,}176$	attention map

Core idea

The input size M affects only the cross-attention read; depth is spent in the compact latent space.

Original Perceiver Output: Pooling Classifier

Fixed-shape output head:

$$L_T = \mathrm{Encoder}(X, L_0)$$

$$z = \frac{1}{N}\sum_{i=1}^{N} L_T[i]$$

$$\mathrm{logits} = zW_{\mathrm{cls}} + b_{\mathrm{cls}}$$

- Mean pooling collapses all latents to one vector
- The classifier predicts one global label
- This is ideal for classification, not dense structured outputs

Where FFN and sharing fit:

- Each latent block is pre-norm attention, residual, FFN, residual
- FFN adds non-linearity after attention mixing
- Weight sharing repeats the processor over iterations

Exam distinction

Perceiver IO replaces pooling with decoder cross-attention driven by output queries.

Computational Complexity Analysis

Model	Self-Attn	Cross-Attn	Total
Standard Transformer	$\mathcal{O}(M^2 d)$	—	$\mathcal{O}(M^2 d \cdot L)$
Perceiver	$\mathcal{O}(N^2 d)$	$\mathcal{O}(MNd)$	$\mathcal{O}((MN + N^2 L)\, d)$

Concrete example (CIFAR-10):

- $M = 1024$ (pixels, 32×32), $N = 96$, $L = 4$
- Standard: $1024^2 = 1{,}048{,}576$
- Perceiver: $1024 \times 96 + 96^2 \times 4 = 135{,}168$
- $\sim 7.8\times$ reduction

Scaling advantage:

- For images 224×224: $M = 50k$
- Transformer: 2.5×10^9 ops
- Perceiver ($N = 512$): $\sim$26M ops
- $\sim 100\times$ reduction

GELU Activation & LAMB Optimizer

GELU (Gaussian Error Linear Unit):

$$\mathrm{GELU}(x) = x \cdot \Phi(x) \approx x \cdot \sigma(1.702\,x)$$

- Smooth activation used in Transformer architectures
- Allows small negative gradient flow

LAMB (Layer-wise Adaptive Moments):

$$r_t^{(l)} = \frac{\|w_t^{(l)}\|}{\|w_t^{(l)} + \eta\,\hat{u}_t^{(l)}\|}$$

Per-layer trust ratio rescales updates and stabilizes large-batch training.

Why LAMB for Perceiver?

- Cross-attention bottleneck creates high variance gradients
- Standard Adam can destabilize with large batches
- LAMB safely scales batch sizes up to 1024

Optimizer	Large Batches	Layer-wise LR
Adam	Diverges	Global
LAMB	**Stable**	**Per-layer trust**

Implementation

Aspect	Paper implementation	Project code
Core model	Cross-attention into learned latents, latent self-attention, FFN, weight sharing	Same modules; weight sharing is exposed as an ablation switch
Model scale	Large vision/language runs: up to 512×1024 latents and 201M-param IO	Smaller runs: CIFAR-10 96×384, ModelNet40/text 128×512
Data setting	ImageNet, ModelNet40, large language pretraining corpora	CIFAR-10, ModelNet40, WikiText-103 byte-level MLM, GLUE fine-tuning
Training budget	Distributed accelerators, batches up to 512–1024	Single-GPU budget, batches 32–128, LAMB retained for stability
Practical additions	Minimal regularization in the reference setup	Dropout, logging, attention-map extraction, and reproducible ablations

The implementation keeps the architectural mechanism intact; the main differences are scale, data budget and training resources.

CIFAR-10: Experimental Setup

Goal: validate the Perceiver on raw pixels without any CNN assumptions.

Hyperparameter	**Value**
Latent dim ($N \times D$)	96×384
Cross-attn stages	4
Transformer blocks	4 (shared)
Attention heads	3
Patch size	2×2
Dropout	0.2
Optimizer	LAMB
Learning rate	0.004
Batch size	64
Epochs	120
Total parameters	**$\sim$3.35M**

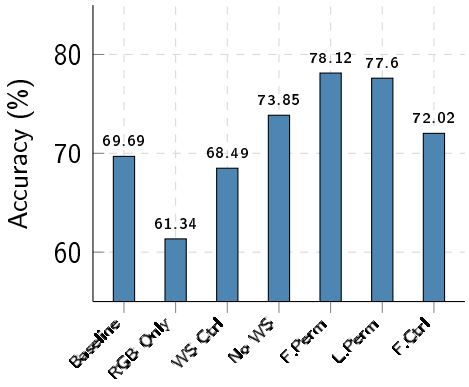
Ablation study plan:

1. Impact of positional encoding
2. Weight sharing vs no sharing
3. Robustness to pixel permutation
4. Best classifier-head configuration

Data pipeline:

- 32×32 RGB images, pixel-level input ($M = 1024$)
- 2D Fourier positional encoding ($K = 64$)
- Standard augmentation: horizontal flip, random crop

CIFAR-10: Ablation Results

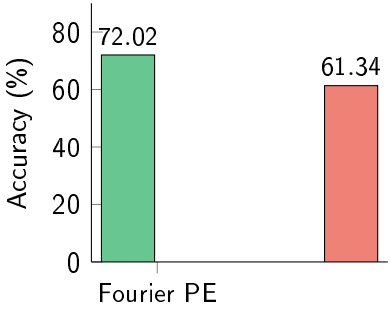


Experiment	Acc.
Fourier Permuted	**78.12%**
Learned PE Permuted	77.60%
No Weight Sharing	73.85%
Fourier Control	72.02%
Baseline Fourier	69.69%
WS Control	68.49%
RGB Only (no PE)	**61.34%**

Key observations:

- PE is essential: -10.68pp without it
- Permuted-pixel runs are best (78.12%), not worst
- Weight sharing: -5.36pp but $\sim 2.6\times$ fewer params

CIFAR-10: Impact of Positional Encoding



Why does this happen?

- Attention is *permutation-equivariant*
- Without PE the model cannot exploit spatial positions
- Color-only signal still classifies, but clearly weaker

Quantitative impact:

- With Fourier PE: **72.02%**
- Without PE (RGB only): **61.34%**
- Delta: -10.68 percentage points
- Still well above chance (10%), but markedly worse

This confirms the original paper's

CIFAR-10: Weight Sharing Analysis

Comparison:

Config	Params	Acc.
Shared (baseline)	3.35M	69.69%
WS Control	3.35M	68.49%
No Sharing	**8.67M**	**73.85%**

Trade-off:

- No sharing: +5.36pp accuracy
- But $\sim$2.6$\times$ more parameters
- Efficiency ratio: $\sim$1.0pp per 1M extra params

Why weight sharing works:

- Acts like a *recurrent* processor in latent space
- Iterative refinement of representations
- Implicit regularization effect
- Critical for deployment on limited hardware

Practical Implication

Weight sharing reaches **92.7%** of the no-sharing accuracy with only **39%** of the parameters — a strong efficiency trade-off for resource-constrained scenarios.

CIFAR-10: Permutation Robustness

Experiment: train *and* evaluate on pixels shuffled by a single fixed permutation σ (same σ throughout).

Configuration	Acc.
Fourier PE (natural order)	69.69%
Fourier PE (permuted)	**78.12%**
Learned PE (permuted)	77.60%

Counterintuitive result:

- Permuting pixels raises accuracy: +8.43pp
- Fourier vs learned (permuted): +0.52pp

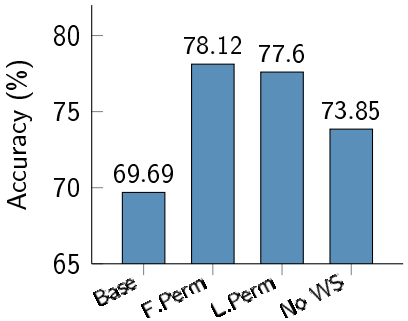
Analysis:

- The Perceiver has **no built-in 2D bias**: position is read only from the PE
- With a fixed σ in train *and* test, it learns the layout from the Fourier features alone
- **Fourier PE** (deterministic) slightly beats learned PE and generalizes better

Insight

Permutation invariance is a **feature**, not a weakness: the model assumes nothing about input ordering, exactly as the paper argues (Jaegle et al., 2021, Tab. 2).

CIFAR-10: Best Perceiver Configuration



Best Perceiver result:

Configuration	Acc.
Fourier PE, permuted	**78.12%**
Learned PE, permuted	77.60%
No weight sharing	73.85%
Baseline Fourier	69.69%

Takeaway:

- Fourier PE carries the spatial signal explicitly.
- Weight sharing is an efficiency–accuracy trade-off.

ModelNet40: Setup & Results

Configuration:

Hyperparameter	Value
Latent dim ($N \times D$)	128×512
Cross-attn stages	2
Transformer blocks	6 (shared)
Points per object	2048
Optimizer	LAMB
Epochs	200
Batch size	128

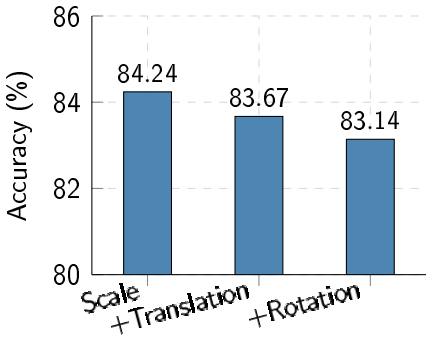
Results:

Configuration	Acc.
Baseline (scale)	**84.24%**
+ Translation	83.67%
+ Rotation	83.14%
Original Paper	*85.7%*

Excellent Reproduction

Gap of only **1.46pp** vs the original paper, despite using batch size 128 vs 512 (GPU constraints).

ModelNet40: Augmentation Effects



Analysis:

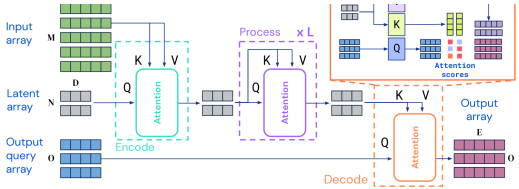
- **Scale only** (baseline): best result at 84.24%
- Adding translation: −0.57pp
- Adding rotation: −1.10pp

Why augmentation doesn't help:

- With 3D Fourier PE the Perceiver already handles rotations natively
- So rotation augmentation is *redundant* — adds noise, not signal
- ModelNet40 objects are already centred, limiting translation's effect

Comparison with paper (85.7%):

Perceiver IO: Structured Outputs



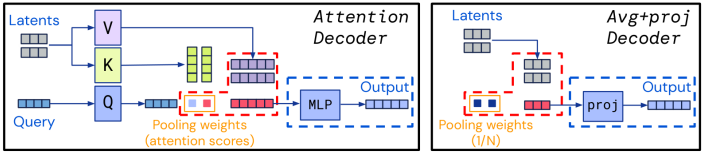
Extension over Perceiver (Jaegle et al., ICLR 2022):

- **Encode**: input $\rightarrow$ latent (same cross-attn encoder)
- **Process**: latent self-attention ($\times L$)
- **Decode**: output queries read latents

Why it matters:

- Original Perceiver: one pooled global output
- IO: task-specific output shape
- Same latent bottleneck, more general decoder

Perceiver IO Decoder with Output Queries



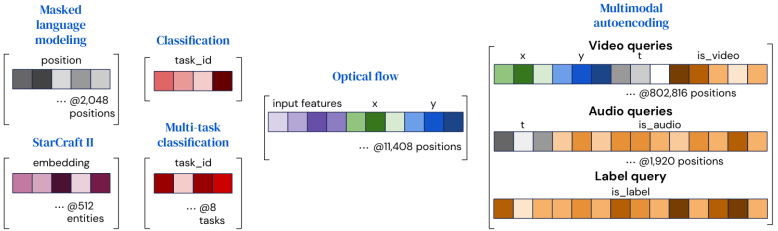
Decoder cross-attention (single layer):

$$Q_{\text{dec}} = \text{OutputQuery}\, W_Q, \quad K_{\text{dec}} = \text{Latent}\, W_K, \quad V_{\text{dec}} = \text{Latent}\, W_V$$

$$\text{Output} = \text{softmax}\left(\frac{Q_{\text{dec}} K_{\text{dec}}^{\top}}{\sqrt{d_k}}\right) V_{\text{dec}} \in \mathbb{R}^{O \times E}$$

- The number of output queries O is **task-dependent** and independent of N and M
- A single decoder cross-attention layer suffices (no self-attention among outputs)

Output Query Design per Task



Classification

One learned query; decoder projection produces class logits.

Dense outputs

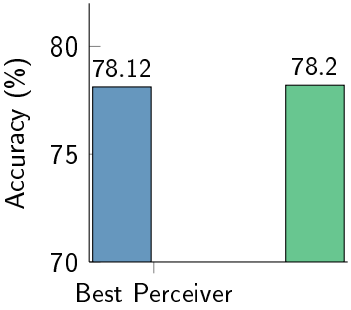
One query per target position, e.g. pixels or voxels.

Masked LM

Masked byte positions query logits over the 256-byte vocabulary.

Output queries define what the decoder reads from the latent array.

CIFAR-10: Perceiver vs Perceiver IO



Why the decoder helps:

- Output query is a learned aggregation instead of fixed mean pooling
- Decoder cross-attention attends selectively to relevant latents
- Small gain on CIFAR-10, but same mechanism enables MLM and dense outputs
- Cost: $\sim 3\times$ parameters (9.5M vs 3.35M)

Model	**Acc.**	**Δ**
Best Perceiver	78.12%	—
Perceiver IO	**78.20%**	**+0.08pp**

WikiText-103: Masked Language Model Pre-training

Goal: pre-train a general language encoder without specialized tokenizers.

Component	Detail
Tokenization	Byte-level (UTF-8)
Vocabulary size	256
Sequence length	1024
Mask probability	15%
Model	Perceiver IO ($\sim$10M)
Epochs / batch	50 / 32
Masked byte acc.	**82.20%**

Why byte-level?

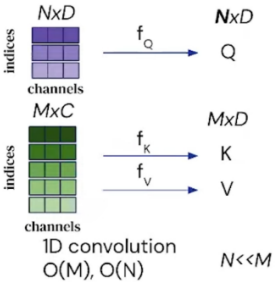
- No BPE/WordPiece tokenizer needed
- Truly language-agnostic input
- Fixed vocabulary of size 256
- Trade-off: longer sequences for same text

MLM task:

- 15% of byte positions masked
- Output queries = masked positions
- Decoder predicts original byte values
- Cross-entropy loss over 256 classes

GLUE: Fine-tuning on 8 Tasks

Transfer pipeline:



Fine-tuning setup:

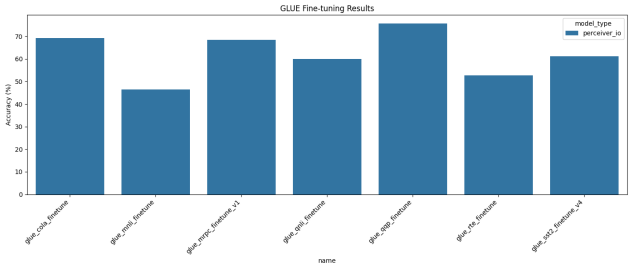
- Transfer pre-trained MLM encoder
- Replace output queries with classification query
- 30 epochs, LR $= 5 \times 10^{-4}$
- Byte-level input (same as pre-training)

8 GLUE benchmark tasks:

- **Single sentence:** CoLA, SST-2
- **Similarity:** MRPC, STS-B, QQP
- **Inference:** MNLI, QNLI, RTE

Each task tests a different NLU capability with the same architecture.

GLUE: Per-Task Analysis



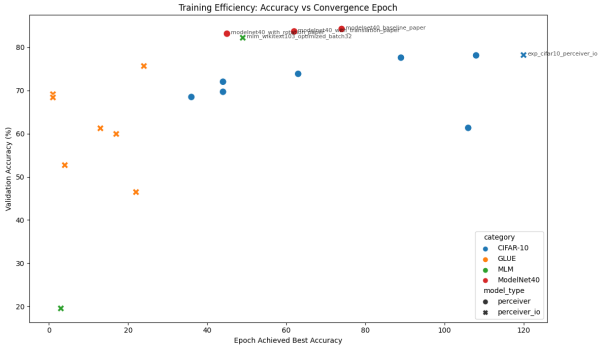
Key observations:

- **QQP** (paraphrase): best task at 75.65%
- **CoLA** 69.13%, **MRPC** 68.38% follow
- **MNLI** (46.47%) & **RTE** (52.71%): hardest — need deep NLI reasoning
- Mean ∼63% vs paper ∼81%

Limitations:

- Byte-level tokenization produces long sequences
- Smaller model (∼10M) vs paper's IO (201M)

Convergence: All Experiments



Convergence speed by modality:

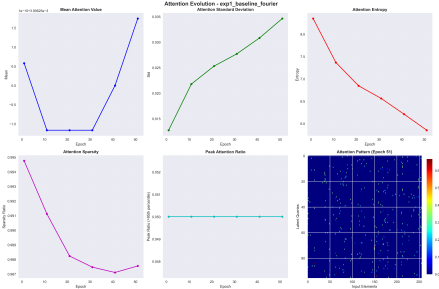
Modality	Epochs
Point Clouds	∼50 (fast)
Text (GLUE FT)	∼15–30
Images (CIFAR-10)	∼120 (slow)

Observations:

- Point clouds: low-dimensional input, fast convergence
- GLUE: transfer learning accelerates training
- CIFAR-10: pixel-level input requires many iterations for spatial learning

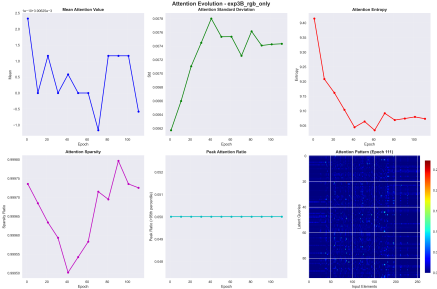
Attention Maps Analysis

Fourier PE (Epoch 81)



Structured, spatially selective patterns (entropy $\sim$6.4). Latents specialize in different image regions.

RGB Only — no PE (Epoch 81)



Diffuse, near-uniform patterns (entropy $\sim$9.1). No spatial selectivity — the model cannot localize information.

Interpretation

Comprehensive Results Summary

Modality	Model	Best Acc.	Paper	Experiments
Images (CIFAR-10)	Perceiver	78.12%	>90%	7 ablations
Images (CIFAR-10)	Perceiver IO	**78.20%**	—	1
Point Clouds (ModelNet40)	Perceiver	84.24%	85.7%	3 augmentations
Text (WikiText-103)	Perceiver IO	82.20% MLM	—	1
NLU (GLUE, 8 tasks)	Perceiver IO	∼63% avg	—	8 fine-tuning

Highlights:

- ModelNet40: only 1.46pp gap from paper
- Perceiver IO slightly edges the best Perceiver
- Successful transfer MLM → GLUE

Overall numbers:

- ∼2500 lines of PyTorch code
- 20+ distinct training runs
- 3 modalities: 2D images, 3D points, text
- 100% reproducible via `reproduce.py`

Gap vs Original Paper

ModelNet40 (Point Clouds):

Source	Acc.
Original Paper	85.7%
Our Implementation	84.24%
Gap	**1.46pp**

Excellent reproduction.
Small gap attributable to reduced batch size (128 vs 512).

CIFAR-10 (Images):

Source	Acc.
Original Paper	>90%
Our Perceiver IO	78.20%
Gap	**∼12pp**

Significant gap.
Caused by hardware constraints: 512 TPU cores vs single consumer GPU.

Root Cause Analysis

The CIFAR-10 gap is primarily due to **batch size** (64 vs 1024), **latent count** (96 vs 256), and **compute time** limitations. The Perceiver architecture is particularly sensitive to these scaling factors on image data.

Hardware Limitations & Design Choices

Parameter	Original Paper (Google)	Our Setup
Hardware	512 TPU v3 cores	1 NVIDIA consumer GPU
Batch size	up to 1024	max 64–128
Latents (N)	256–512	96–128
Model size	30–100M+ params	3.35–10M params
Training time	days (distributed)	hours–days (single GPU)

Consequences:

- LAMB optimizer less effective with small batches
- Fewer latents limit representational capacity
- Reduced augmentation due to time constraints
- Fewer hyperparameter search iterations

Mitigation strategies:

- Weight sharing to reduce memory
- Gradient accumulation where possible
- Careful cosine annealing schedule
- Focus experiments on most informative ablations

Possible Improvements

Architecture:

- Increase latent count ($N = 256+$) with gradient checkpointing
- Add multi-scale cross-attention (coarse $\rightarrow$ fine)
- Explore relative positional encodings (RoPE, ALiBi)
- Deeper decoder for structured outputs

Training:

- Larger batch sizes with gradient accumulation
- Mixed precision (FP16/BF16) training
- Longer training schedules

Data & Tasks:

- ImageNet pre-training for vision
- Subword tokenization for NLP
- Multi-modal joint training
- Audio modality experiments

Analysis:

- Layer-wise attention probing
- Latent space clustering visualization
- Systematic hyperparameter search
- Comparison with efficient Transformers (Linformer, Performer)

Lessons Learned

What works well:

- Genuinely modality-agnostic architecture
- Weight sharing: $\sim 2.6\times$ fewer parameters, modest accuracy cost
- Perceiver IO slightly edges base Perceiver on CIFAR-10
- Transfer learning MLM $\rightarrow$ GLUE is effective
- Fourier PE is robust and general-purpose

Observed limitations:

- Performance gap vs paper on images ($\sim$12pp)
- LAMB optimizer sensitive to batch size
- Byte-level tokenization produces very long sequences for text
- Augmentation can hurt with limited training
- Hyperparameter tuning is time-consuming

Main Takeaway

Positional encoding is the single most critical component of the architecture: without it, accuracy drops from 72% to 61% on CIFAR-10, confirming the paper's core hypothesis.

Conclusions

Strengths of the Perceiver:

- Single architecture for images, point clouds, and text
- Linear complexity in input size
- Strong memory efficiency via weight sharing ($\sim$61%)
- Zero CNN/RNN domain-specific assumptions
- Output queries enable flexible structured decoding

Weaknesses:

- Requires large compute to match specialized models
- Heavily depends on positional encoding
- Outperformed by efficient CNNs on standard vision benchmarks
- LAMB optimizer introduces training instability at small scale
- Latent bottleneck limits fine-grained spatial reasoning

Final Assessment

The Perceiver demonstrates that a **single general-purpose architecture** can handle diverse modalities. While it does not yet match domain-specific models at limited scale, the paradigm

References

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Appendix: Expected Questions

Architecture:

1. Why cross-attention instead of full self-attention?
   → Reduces $\mathcal{O}(M^2)$ to $\mathcal{O}(NM)$
2. How does weight sharing affect performance?
   → -5.36pp for $\sim 2.6\times$ fewer params
3. Why Fourier PE over learned PE?
   → More robust to permutation, generalizes better
4. What is the role of GEGLU?
   → Higher non-linear capacity, stable gradients

Experiments:

5. Why is the CIFAR-10 gap so large?
   → Hardware limits: batch 64 vs 1024, 96 vs 256 latents
6. Why byte-level tokenization for NLP?
   → Truly modality-agnostic, no external tokenizer
7. How does Perceiver IO improve on Perceiver?
   → Learned output queries vs mean pooling
8. Could this replace ViT/BERT?
   → Not yet at limited scale; promising at Google scale

Thank you!

Questions?

Francesco Gigli e Corso Vignoli

Code and models available upon request

Code:	∼2500 lines PyTorch (from-scratch)
Experiments:	20+ training runs, 3 modalities
Reproducibility:	`python reproduce.py`